



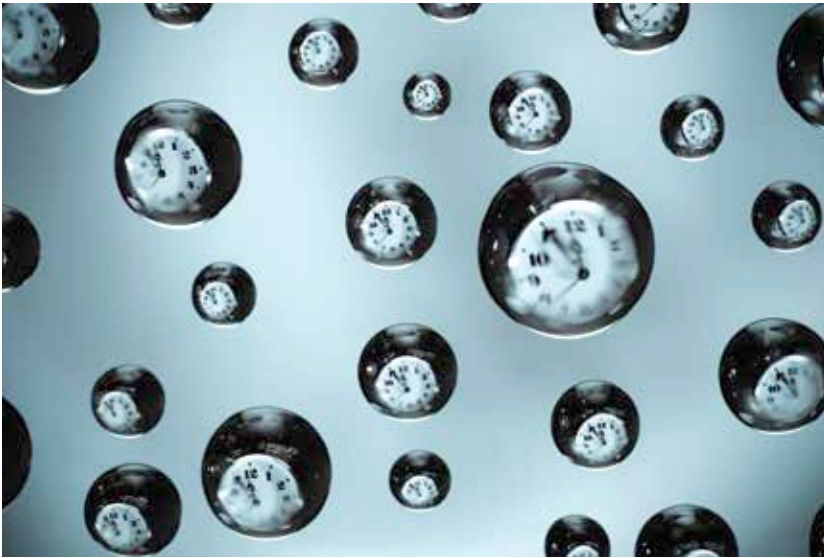
The flamingo's stance

The flamingo prefers to stand on one leg due to the minimal muscle effort required, according to a study conducted recently <http://digit.in/1legstand>



Murder inciting play?

A Polish play named *Klątwa* or *The Curse* has been accused by prosecutors of inciting the audience to murder. <http://digit.in/TheCurz>



The mysteries of Time Travel

The first thing you need to know about time travel is that it's possible. The second thing you need to know about time travel is that it's already been done...

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Russian cosmonaut Sergei Krikalev has spent over 803 days on the ISS, orbiting the earth at 7.66 km/s, and as a result, he arrived on Earth 0.02 seconds into the future. Of course, it may not meet the standards of science fiction movies like *Back to the Future* where you hit the button on a flux capacitor and land up at your dad's prom night, but for all intents and purposes, it is time travel.

Before we figure out how Sergei landed into the future, and how you've done the same countless times in your life, there's a more fundamental question that we need to tackle.

WHAT IS TIME ANYWAY?

The best answer for this possibly comes from Einstein, who after a lot of thought, simply said: "Time is what the clock measures."

This is the generally accepted definition of time in physics. Now, it may sound like an oversimplification, but it conveys an important truth: Time is a measureable and relative quantity. This is how Sergei leapt into the future, by travelling at high speeds for a long time, time slowed down for him relative to an observer on earth. In other words, he experienced time dilation. You've experienced time dilation as well, the last time you took a flight, drove around in your car or even you went for a jog – the only difference is that time dilation in those cases is insignificant.

This stretchy picture of time may seem counter intuitive, but that's only because we humans perceive time as a standard, linear quantity. In reality, time slows down, speeds up, and even bends over itself forming a loop, depending on the location of the observer. While working on the General Theory of Relativity, Einstein coined the term spacetime to depict how time is tied in with the three dimensions of space to form a four-dimensional "fabric" – on this fabric lie all gravitational objects such as planets and stars. Gravity, in fact, bends and distorts this fabric, the stronger the gravity, the bigger the distortion in spacetime.

In his book *Quantum Theory Cannot Hurt You*, Marcus Chown states that you can only go back in time so far as when the first time machine was invented, anything before that is unrecorded history. This explains why we aren't overrun by tourists from the future – doesn't mean time travel is not possible – just that they can't reach us in a time before we invent it.

To us, time always flows from past to present and into the future. Scientists call this the arrow of time. Nature makes going back in time difficult in comparison to moving forward, if not impossible. First, you have to travel faster than the speed of light to travel back in time, but doing so is prohibited by the laws of physics. Another reason that past time travel is trickier is because entropy increases with time, i.e., things move towards more disorderly states. You see a teacup fall and break,



Time is the fourth dimension woven into the three dimensions of space to form the spacetime continuum which permeates every corner of the universe